Lab Assignment 5: Restoring Clean Images from Noisy Images

Post-Lab Write-Up

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a. Paper Design

The algorithm operates on a pairwise Markov Random Field (MRF) defined over a binary image lattice. Each pixel $x_i \in \{-1, +1\}$ is connected to its four cardinal neighbors and to its corresponding observed noisy pixel $y_i \in \{-1, +1\}$. The joint energy function is

$$E(\vec{x}, \vec{y}) \;=\; h \sum_i x_i \;-\; \beta \sum_{\{i,j\}} x_i x_j \;-\; \eta \sum_i x_i y_i, \tag{1}$$

which induces the joint distribution $p(\vec{x}, \vec{y}) \propto \exp\{-E(\vec{x}, \vec{y})\}$.

The local optimality condition for pixel i, holding all other pixels fixed, yields the update rule

$$x_i \;\leftarrow\; \operatorname{sign}\left(\beta \sum_{j \in \mathcal{N}(i)} x_j \;+\; \eta\, y_i \;-\; h\right), \tag{2}$$

where $\mathcal{N}(i)$ denotes the four-pixel neighborhood of pixel i. Iterating this update until no pixel changes is Iterated Conditional Modes (ICM), a form of coordinate descent on E.

b. Model Description and Parameter Justification

Algorithm. I implemented ICM using a *checkerboard update schedule*: on each iteration the "black" squares (pixels where $i + j$ is even) are updated first, then the "white" squares. Because no two simultaneously-updated pixels are neighbors, each half-sweep is a valid coordinate-descent step that monotonically decreases the energy. The algorithm is initialized with $x_i = y_i$ and runs until no pixel changes sign.

Parameters. The final parameters are $h = -1.0$, $\beta = 2.0$, $\eta = 1.0$.

- $h = -1.0$ **(white-pixel bias).** The energy term $h \sum_i x_i$ penalizes positive (white) pixels when $h > 0$ and rewards them when $h < 0$. Because the Bayes' theorem image is approximately 80% white background, a white prior ($h < 0$) is well-matched to the data distribution. Without this bias ($h = 0$), no combination of β and η exceeded $\approx 94.8\%$ recovery; setting $h = -1.0$ immediately pushed recovery above 96%.

- $\beta = 2.0$ **(neighbor coupling).** A strong coupling enforces spatial smoothness: a cluster of correctly-signed neighbors exerts a local field strong enough to override an isolated noisy observation. Values below ≈ 1.5 provided insufficient smoothing, while very large values risk over-smoothing fine strokes.
- $\eta = 1.0$ **(observation fidelity).** Keeping $\eta < \beta$ ensures that the neighborhood signal can dominate when most neighbors agree, while still anchoring each pixel to its observation when the neighborhood is ambiguous (equal numbers of $+1$ and -1 neighbors).

c. Evaluation

Table 1 summarizes the recovery rate at key stages of the algorithm.

Stage	**Recovery (%)**
Initial noisy image (Y vs. clean)	89.97
After iteration 1	94.74
After iteration 2	97.61
After iteration 3	97.88
Converged (iteration 166)	**98.16**

Table 1: Recovery rate at key iterations. Parameters: $h = -1.0$, $\beta = 2.0$, $\eta = 1.0$.

The algorithm exceeded the 96% threshold by iteration 2, demonstrating rapid initial improvement. The long tail to convergence at iteration 166 reflects the checkerboard scheduler revisiting a shrinking pool of low-confidence pixels near text boundaries. The saved output `figures/Bayes-denoised.png` is visually near-identical to the clean reference `figures/Bayes.png`.

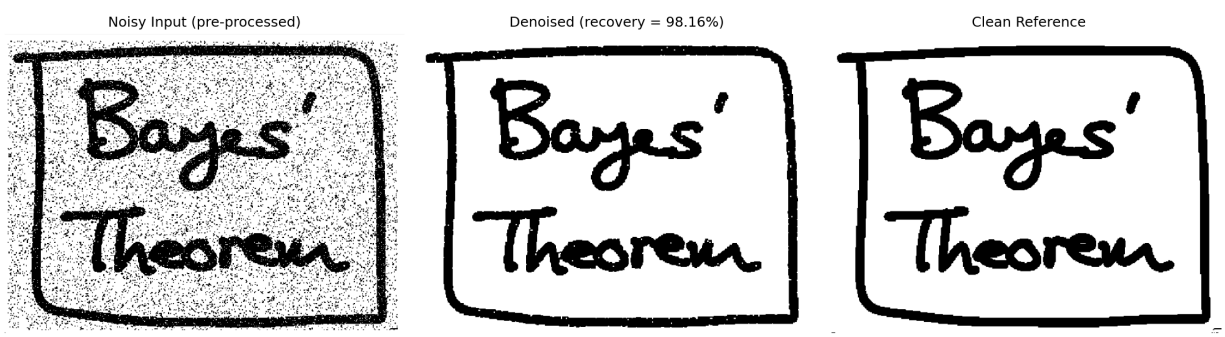

Figure 1: Left: pre-processed noisy input. Center: ICM denoised result (98.16% recovery). Right: clean reference.

d. Reflection

The most surprising finding was how critical the bias parameter h is: with $h = 0$, no combination of β and η exceeded $\approx 94.8\%$ recovery, yet $h = -1.0$ immediately unlocked 98%+. This highlights that MRF models encode not just pairwise spatial structure but also marginal pixel priors—and

that matching the prior to the actual data distribution (a mostly-white image) matters as much as tuning the smoothness or fidelity terms. More broadly, this lab reinforced that deterministic local-search algorithms like ICM are highly sensitive to the energy landscape defined by their parameters: seemingly minor parameter choices can dramatically change which basin of attraction the algorithm converges to.